

Algebra: Quadratic Formula & Completing the Square

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DIRECTIONS

Solve by completing the square or using the Quadratic Formula. Simplify all answers.

1 Complete the square — write in vertex form:

$$x^2 + 2x + 1$$

Answer: _____

2 Complete the square — write in vertex form:

$$x^2 + 2x - 6$$

Answer: _____

3 Complete the square — write in vertex form:

$$x^2 + 2x - 7$$

Answer: _____

4 Complete the square — write in vertex form:

$$x^2 + 4x - 2$$

Answer: _____

5 Solve using the Quadratic Formula:

$$1x^2 + 4x + 5 = 0$$

Answer: _____

6 Solve using the Quadratic Formula:

$$1x^2 - 4x + 5 = 0$$

Answer: _____

7 Solve using the Quadratic Formula:

$$1x^2 + 5x + 2 = 0$$

Answer: _____

8 Solve using the Quadratic Formula:

$$2x^2 - 4x - 2 = 0$$

Answer: _____

9 Solve using the Quadratic Formula:

$$1x^2 - 5x - 2 = 0$$

Answer: _____

10 Solve using the Quadratic Formula:

$$2x^2 - 4x - 2 = 0$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Completing the square: half of b , then square it. QF: $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$.

1 Complete the square — write in vertex form:

$$= (x + 1)^2 + 0$$

$$x^2 + 2x + 1$$

2 Complete the square — write in vertex form:

$$= (x + 1)^2 - 7$$

$$x^2 + 2x - 6$$

3 Complete the square — write in vertex form:

$$= (x + 1)^2 - 8$$

$$x^2 + 2x - 7$$

4 Complete the square — write in vertex form:

$$= (x + 2)^2 - 6$$

$$x^2 + 4x - 2$$

5 Solve using the Quadratic Formula:

$$= \text{No real solutions}$$

$$1x^2 + 4x + 5 = 0$$

6 Solve using the Quadratic Formula:

$$= \text{No real solutions}$$

$$1x^2 - 4x + 5 = 0$$

7 Solve using the Quadratic Formula:

$$= x = -\frac{5}{2} - \frac{\sqrt{17}}{2}, x = -\frac{5}{2} + \frac{\sqrt{17}}{2}$$

$$1x^2 + 5x + 2 = 0$$

8 Solve using the Quadratic Formula:

$$= x = 1 - \sqrt{2}, x = 1 + \sqrt{2}$$

$$2x^2 - 4x - 2 = 0$$

9 Solve using the Quadratic Formula:

$$= x = \frac{5}{2} - \frac{\sqrt{33}}{2}, x = \frac{5}{2} + \frac{\sqrt{33}}{2}$$

$$1x^2 - 5x - 2 = 0$$

10 Solve using the Quadratic Formula:

$$= x = 1 - \sqrt{2}, x = 1 + \sqrt{2}$$

$$2x^2 - 4x - 2 = 0$$



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